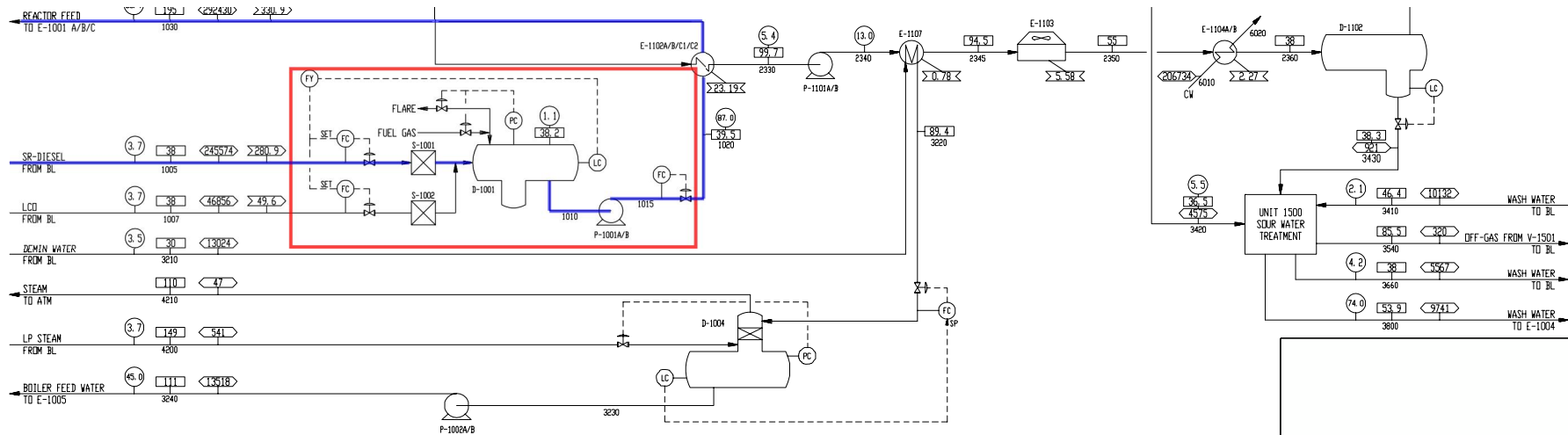


S-1001
SR FEED FILTER

S-1002
LCD FEED FILTER

P-1001A/B
FEED PUMP
DESIGN CAPACITY: 370 m³/h
HEAD: 1192 m

○ PRESSURE KG/CMG
□ TEMPERATURE °C
▭ FLOW MMG/H (< °C: 760 MMHG)
▭ FLOW KG/H
▭ DUTY GCM/H
▭ FLOW MMG/H



Att-1 (1/3)

(Partial)

C-1201

RECYCLE GAS COMPRESSOR
PACKAGE TO BE DETAILED BY VENDOR
DESIGN CAPACITY: 146633 Nm³/h

E-1002

PROCESS BOILER
DESIGN DUTY: 6.89 Gcal/h

R-1001

HDS REACTOR
TL: 23460 mm
ID: 4500 mm

E-1001 A/B/C

REACTOR FEED/EFFLUENT EXCHANGER
DESIGN DUTY: 41.01 Gcal/h

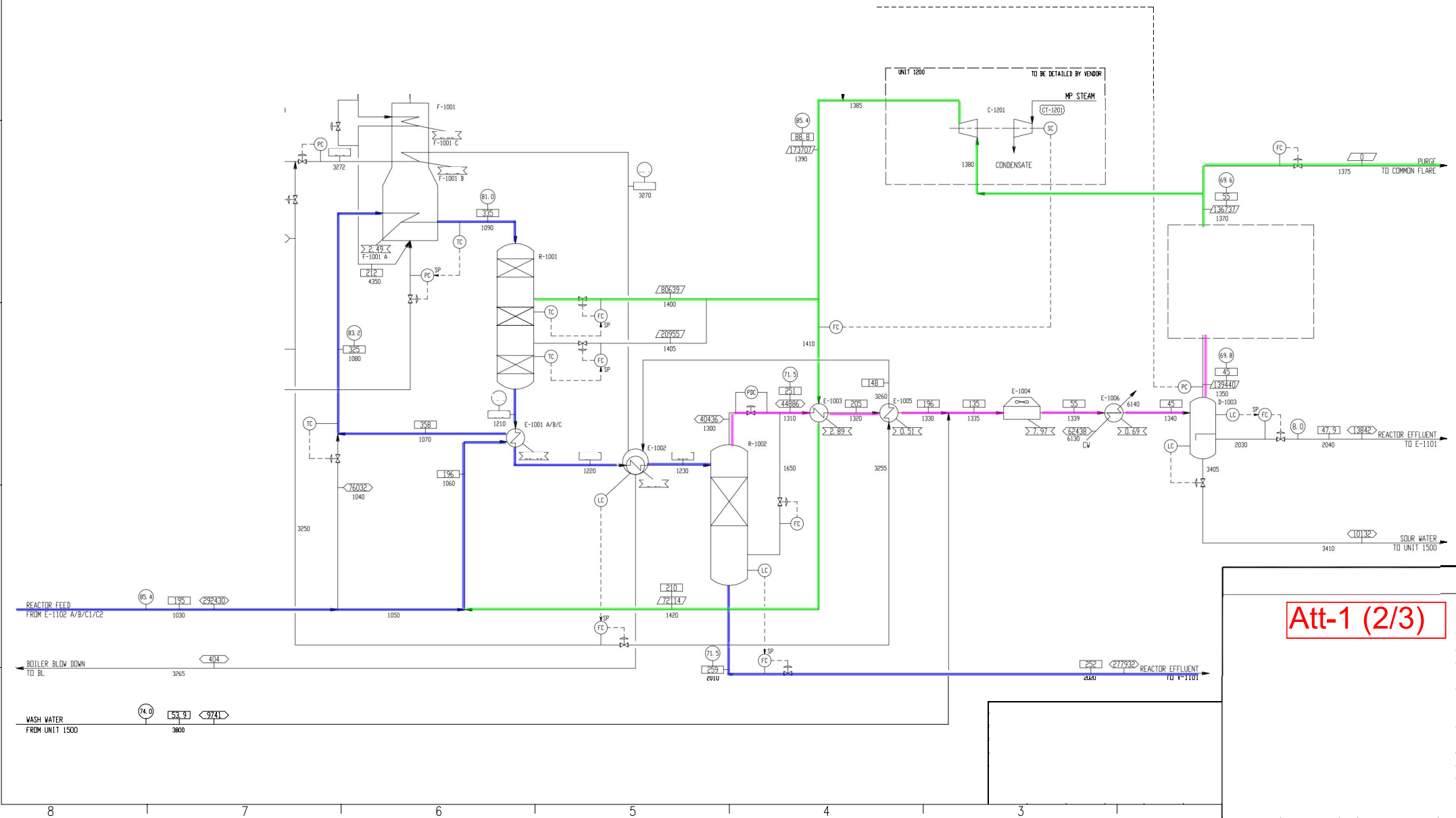
F-1001

REACTOR FEED HEATER
DESIGN ABSORBED DUTY: 6.09 Gcal/h

R-1002

POLYSHIFT REACTOR/HOT SEPARATOR
TL: 19000 mm
ID: 3000 mm

○ PRESSURE KG/CMG
□ TEMPERATURE °C
▬ FLOW NM³/H (O °C, 760 MMHG)
▬ FLOW KG/H
▬ DUTY GCA/H
▬ FLOW SKE/H



Att-1 (2/3)

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A	<table><tr><td>STREAM NO.</td><td>1005</td><td>1007</td><td>1010</td><td>1015</td></tr><tr><td>STREAM DESCRIPTION</td><td>SR</td><td>LCO</td><td>P-1001</td><td>P-1001</td></tr><tr><td></td><td>FEED</td><td>FEED</td><td>SUCTION</td><td>DISCHARGE</td></tr><tr><td>PHASE</td><td>LIQUID</td><td>LIQUID</td><td>LIQUID</td><td>LIQUID</td></tr><tr><td>TOTAL MOLAR RATE (KG-MOLE/H)</td><td>938.79</td><td>239.76</td><td>1179.95</td><td>1179.95</td></tr><tr><td>TOTAL MASS FLOW (KG/H)</td><td>245574</td><td>46866</td><td>292430</td><td>292430</td></tr><tr><td>TEMPERATURE (°C)</td><td>38.0</td><td>38.0</td><td>38.2</td><td>39.3</td></tr><tr><td>PRESSURE (KG/CM2 G)</td><td>3.7</td><td>3.7</td><td>1.1</td><td>90.4</td></tr><tr><td>TOTAL ENTHALPY (M*KCAL/H)</td><td>2.866</td><td>0.531</td><td>3.408</td><td>4.288</td></tr><tr><td></td><td>(KCAL/KG)</td><td>11.673</td><td>11.335</td><td>11.654</td></tr><tr><td>MOLE FRACTION LIQUID</td><td>1.000</td><td>1.000</td><td>1.000</td><td>1.000</td></tr><tr><td>SPECIFIC GRAVITY</td><td>0.8750</td><td>0.9460</td><td>0.8847</td><td>0.8847</td></tr><tr><td>VAPOR RATE (KG/H)</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>STD. VAPOR RATE (NM3/H)</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>VAPOR MOLECULAR WEIGHT</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>VAPOR ENTHALPY (KCAL/KG)</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>VAPOR CP (KCAL/KG-C)</td><td>0.000</td><td>0.000</td><td>0.000</td><td>0.000</td></tr><tr><td>VAPOR DENSITY (KG/M3)</td><td>0.000</td><td>0.000</td><td>0.000</td><td>0.000</td></tr><tr><td>VAPOR THERMAL CONDUCTIVITY (KCAL/HM-C)</td><td>0.0000</td><td>0.0000</td><td>0.0000</td><td>0.0000</td></tr><tr><td>VAPOR VISCOSITY (CP)</td><td>0.0000</td><td>0.0000</td><td>0.0000</td><td>0.0000</td></tr><tr><td>LIQUID RATE (M3/H)</td><td>285.24</td><td>50.36</td><td>336.19</td><td>332.63</td></tr><tr><td>STD. LIQUID RATE (M3/H)</td><td>280.93</td><td>49.58</td><td>330.87</td><td>330.87</td></tr><tr><td>LIQUID MOLECULAR WEIGHT</td><td>261.59</td><td>195.43</td><td>247.83</td><td>247.83</td></tr><tr><td>LIQUID ENTHALPY (KCAL/KG)</td><td>11.673</td><td>11.335</td><td>11.654</td><td>14.662</td></tr><tr><td>LIQUID CP (KCAL/KG-C)</td><td>0.4475</td><td>0.3980</td><td>0.4405</td><td>0.4402</td></tr><tr><td>LIQUID DENSITY (KG/M3)</td><td>893.94</td><td>930.43</td><td>869.87</td><td>873.13</td></tr><tr><td>LIQUID SURFACE TENSION (DYNE/CM)</td><td>31.100</td><td>34.124</td><td>31.416</td><td>31.321</td></tr><tr><td>LIQUID THERMAL CONDUCTIVITY (KCAL/HM-C)</td><td>0.0914</td><td>0.0969</td><td>0.0920</td><td>0.0953</td></tr><tr><td>LIQUID VISCOSITY (CP)</td><td>5.4329</td><td>3.5976</td><td>4.9049</td><td>5.5445</td></tr><tr><td>COMPOSITION MOLAR RATE (KG-MOLE/H)</td><td></td><td></td><td></td><td></td></tr><tr><td>H2O</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>NH3</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>H2S</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>H2</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>METHANE</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>ETHANE</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>PROPANE</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>BUTANE</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>PENTANE</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>DIPA</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>AIR/FLUEGAS</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>NAPHTHA(150)</td><td>0.00</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>DISEL(150+)</td><td>938.79</td><td>239.76</td><td>1179.95</td><td>1179.95</td></tr></table>								STREAM NO.	1005	1007	1010	1015	STREAM DESCRIPTION	SR	LCO	P-1001	P-1001		FEED	FEED	SUCTION	DISCHARGE	PHASE	LIQUID	LIQUID	LIQUID	LIQUID	TOTAL MOLAR RATE (KG-MOLE/H)	938.79	239.76	1179.95	1179.95	TOTAL MASS FLOW (KG/H)	245574	46866	292430	292430	TEMPERATURE (°C)	38.0	38.0	38.2	39.3	PRESSURE (KG/CM2 G)	3.7	3.7	1.1	90.4	TOTAL ENTHALPY (M*KCAL/H)	2.866	0.531	3.408	4.288		(KCAL/KG)	11.673	11.335	11.654	MOLE FRACTION LIQUID	1.000	1.000	1.000	1.000	SPECIFIC GRAVITY	0.8750	0.9460	0.8847	0.8847	VAPOR RATE (KG/H)	0	0	0	0	STD. VAPOR RATE (NM3/H)	0	0	0	0	VAPOR MOLECULAR WEIGHT	0.00	0.00	0.00	0.00	VAPOR ENTHALPY (KCAL/KG)	0.00	0.00	0.00	0.00	VAPOR CP (KCAL/KG-C)	0.000	0.000	0.000	0.000	VAPOR DENSITY (KG/M3)	0.000	0.000	0.000	0.000	VAPOR THERMAL CONDUCTIVITY (KCAL/HM-C)	0.0000	0.0000	0.0000	0.0000	VAPOR VISCOSITY (CP)	0.0000	0.0000	0.0000	0.0000	LIQUID RATE (M3/H)	285.24	50.36	336.19	332.63	STD. LIQUID RATE (M3/H)	280.93	49.58	330.87	330.87	LIQUID MOLECULAR WEIGHT	261.59	195.43	247.83	247.83	LIQUID ENTHALPY (KCAL/KG)	11.673	11.335	11.654	14.662	LIQUID CP (KCAL/KG-C)	0.4475	0.3980	0.4405	0.4402	LIQUID DENSITY (KG/M3)	893.94	930.43	869.87	873.13	LIQUID SURFACE TENSION (DYNE/CM)	31.100	34.124	31.416	31.321	LIQUID THERMAL CONDUCTIVITY (KCAL/HM-C)	0.0914	0.0969	0.0920	0.0953	LIQUID VISCOSITY (CP)	5.4329	3.5976	4.9049	5.5445	COMPOSITION MOLAR RATE (KG-MOLE/H)					H2O	0.00	0.00	0.00	0.00	NH3	0.00	0.00	0.00	0.00	H2S	0.00	0.00	0.00	0.00	H2	0.00	0.00	0.00	0.00	METHANE	0.00	0.00	0.00	0.00	ETHANE	0.00	0.00	0.00	0.00	PROPANE	0.00	0.00	0.00	0.00	BUTANE	0.00	0.00	0.00	0.00	PENTANE	0.00	0.00	0.00	0.00	DIPA	0.00	0.00	0.00	0.00	AIR/FLUEGAS	0.00	0.00	0.00	0.00	NAPHTHA(150)	0.00	0.00	0.00	0.00	DISEL(150+)	938.79	239.76	1179.95	1179.95
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C	<table><tr><td>STREAM NO.</td><td>1340</td><td>1360</td><td>1370</td><td>1375</td></tr><tr><td>STREAM DESCRIPTION</td><td>D-1003</td><td>V-1401</td><td>V-1401</td><td>PURGE</td></tr><tr><td></td><td>INLET</td><td>GAS INLET</td><td>GAS OUTLET</td><td></td></tr><tr><td>PHASE</td><td>MIKED</td><td>VAPOR</td><td>VAPOR</td><td>VAPOR</td></tr><tr><td>TOTAL MOLAR RATE (KG-MOLE/H)</td><td>6856.86</td><td>6221.09</td><td>6100.50</td><td>0.00</td></tr><tr><td>TOTAL MASS FLOW (KG/H)</td><td>54786</td><td>31128</td><td>27113</td><td>0</td></tr><tr><td>TEMPERATURE (°C)</td><td>45.0</td><td>45.0</td><td>55.0</td><td>55.0</td></tr><tr><td>PRESSURE (KG/CM2 G)</td><td>69.8</td><td>69.8</td><td>69.6</td><td>69.6</td></tr><tr><td>TOTAL ENTHALPY (M*KCAL/H)</td><td>6.659</td><td>5.941</td><td>5.899</td><td>0.000</td></tr><tr><td></td><td>(KCAL/KG)</td><td>121.553</td><td>190.851</td><td>217.556</td></tr><tr><td>MOLE FRACTION LIQUID</td><td>0.062</td><td>0.000</td><td>0.000</td><td>0.000</td></tr><tr><td>SPECIFIC GRAVITY</td><td>0.2458</td><td>0.1587</td><td>0.1421</td><td>0.1421</td></tr><tr><td>VAPOR RATE (KG/H)</td><td>31241</td><td>31128</td><td>27113</td><td>0</td></tr><tr><td>STD. VAPOR RATE (NM3/H)</td><td>139586</td><td>139440</td><td>136736</td><td>0</td></tr><tr><td>VAPOR MOLECULAR WEIGHT</td><td>5.02</td><td>5.00</td><td>4.44</td><td>4.44</td></tr><tr><td>VAPOR ENTHALPY (KCAL/KG)</td><td>190.22</td><td>190.85</td><td>217.56</td><td>217.56</td></tr><tr><td>VAPOR CP (KCAL/KG-C)</td><td>1.472</td><td>1.477</td><td>1.655</td><td>1.655</td></tr><tr><td>VAPOR DENSITY (KG/M3)</td><td>12.791</td><td>12.638</td><td>10.830</td><td>10.830</td></tr><tr><td>VAPOR THERMAL CONDUCTIVITY (KCAL/HM-C)</td><td>0.0865</td><td>0.0869</td><td>0.0948</td><td>0.0948</td></tr><tr><td>VAPOR VISCOSITY (CP)</td><td>0.0115</td><td>0.0115</td><td>0.0113</td><td>0.0113</td></tr><tr><td>LIQUID RATE (M3/H)</td><td>27.29</td><td>0.00</td><td>0.00</td><td>0.00</td></tr><tr><td>STD. 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VAPOR RATE (NM3/H)	139586	139440	136736	0	VAPOR MOLECULAR WEIGHT	5.02	5.00	4.44	4.44	VAPOR ENTHALPY (KCAL/KG)	190.22	190.85	217.56	217.56	VAPOR CP (KCAL/KG-C)	1.472	1.477	1.655	1.655	VAPOR DENSITY (KG/M3)	12.791	12.638	10.830	10.830	VAPOR THERMAL CONDUCTIVITY (KCAL/HM-C)	0.0865	0.0869	0.0948	0.0948	VAPOR VISCOSITY (CP)	0.0115	0.0115	0.0113	0.0113	LIQUID RATE (M3/H)	27.29	0.00	0.00	0.00	STD. LIQUID RATE (M3/H)	26.74	0.00	0.00	0.00	LIQUID MOLECULAR WEIGHT	37.42	0.00	0.00	0.00	LIQUID ENTHALPY (KCAL/KG)	30.443	0.000	0.000	0.000	LIQUID CP (KCAL/KG-C)	0.6327	0.0000	0.0000	0.0000	LIQUID DENSITY (KG/M3)	863.83	0.00	0.00	0.00	LIQUID SURFACE TENSION (DYNE/CM)	29.021	0.000	0.000	0.000	LIQUID THERMAL CONDUCTIVITY (KCAL/HM-C)	0.1703	0.0000	0.0000	0.0000	LIQUID VISCOSITY (CP)	0.5166	0.0000	0.0000	0.0000	COMPOSITION MOLAR RATE (KG-MOLE/H)					H2O	552.09	4.71	4.71	0.00	NH3	4.41	0.02	0.02	0.00	H2S	119.50	113.46	0.32	0.00	H2	5449.28	5445.41	5444.00	0.00	METHANE	292.40	292.77	290.00	0.00	ETHANE	61.87	61.28	60.40	0.00	PROPANE	34.00	32.99	32.00	0.00	BUTANE	15.55	13.77	13.49	0.00	PENTANE	0.00	0.00	0.00	0.00	DIPA	0.00	0.00	0.00	0.00	AIR/FLUEGAS	252.97	253.53	252.00	0.00	NAPHTHA(150)	10.25	3.10	3.10	0.00	DISEL(150+)	64.53	0.45	0.45	0.00
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